

World of microbes and their impact on oral health

OMFS 513: Integrated Biomedical and Clinical Sciences 2

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Objectives & Competency Statements

Learning Outcomes

1. Discuss the oral and systemic manifestations associated with selected bacterial infections not associated with dental caries and periodontal disease.
2. Briefly review the management of the bacterial infections.
3. Begin the process of creating differential diagnosis.

Competency statements met

1.3, 2.1, 3.1, 4.4, and 5.1

Streptococcal Pharyngitis

- Group A, beta hemolytic streptococci.
- Symptoms:
 - Fever, malaise, sore throat with yellow exudate on the tonsils, and erythema of the oropharynx
 - Cervical lymphadenopathy
- Treatment
 - Self-limiting
 - Support with antibiotics, fluids, and rest
- Complications
 - Rheumatic fever (rare)
 - Risk factor for infective endocarditis



Britannica, The Editors of Encyclopaedia. "tonsillitis". *Encyclopedia Britannica*, 18 Feb. 2023, <https://www.britannica.com/science/tonsillitis>.

Scarlet Fever

- Caused by Group A, beta-hemolytic Streptococci
- Begins as a tonsillitis with pharyngitis
- Streptococci release a toxin that attacks blood vessels causing a skin rash
- Fever
- Rash develops within two days and spreads within 24 hours

Scarlet Fever

- Skin
 - Rash “sunburn with goose pimples”
 - Pastia’s line-transverse red streaks occurring in the skin folds
 - Red cheeks and circumoral pallor



Tatiana Ninkov and Mike Cadogan, "Second disease," In: LITFL - Life in the FastLane, Accessed on March 30, 2023.

Scarlet Fever- Strawberry Tongue



White strawberry tongue - white coating with hyperplastic fungiform papillae



Red strawberry tongue – red discoloration with hyperplastic fungiform papillae

Tatiana Ninkov and Mike Cadogan, "Second disease," In: LITFL - Life in the FastLane, Accessed on March 30, 2023.

Tonsillolithiasis

- Occurs when an accumulation of bacteria and debris in the tonsillar crypts becomes calcified.
- Symptoms
 - Enlargement of the tonsillar tissue
 - Pain
 - Halitosis
- Treatment
 - Removal of stones (gargling, pulsating water jets)
 - Surgical removal of tonsils

Tonsillolithiasis



<https://www.youtube.com/watch?v=PC2rVGyYg0o>

Tonsilolithiasis



B.,B.B.,M.L.,A.T.,Avinash,C.A.,B.,C.(2013).*Tonsillolith: A Panoramic Radiograph Presentation*,J Clin of Diagn Res. 7(10), 2378-2379.

Cat-Scratch Disease

- Cause bacterial organism *Bartonella henselae*
- Most common cause of chronic regional lymphadenopathy in children
- Occurs after contact (scratches, bites, and licks) from a cat
- Papules and pustules develop along the scratch line 3-14 days after the injury
- Lymphadenopathy, fever or malaise

Cat-Scratch Disease

- Preauricular and cervical lymph nodes enlargement
- Skin rashes
- Systemic changes
 - Organ enlargement
 - Thrombocytopenia



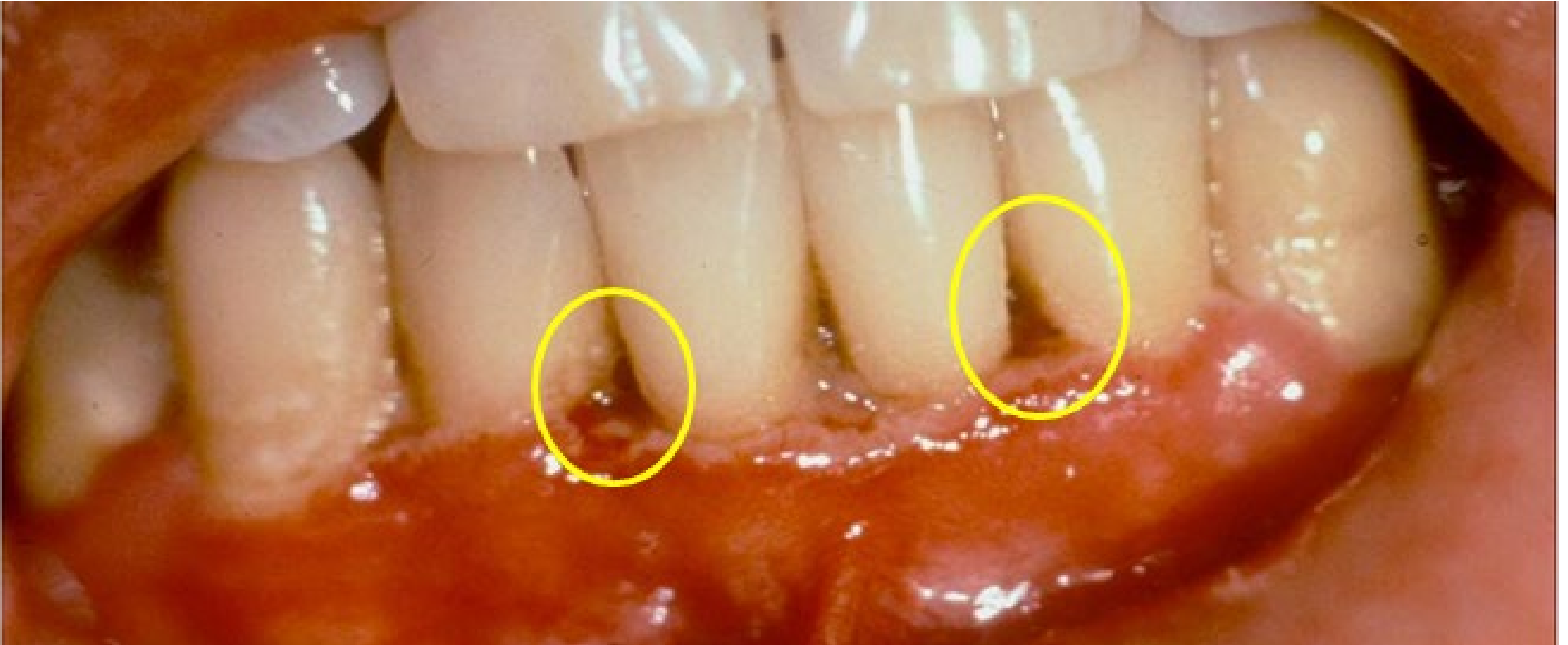
Figure 1. Lieberman J. Cancer Therapy Advisor.
cancertherapyadvisor.com/home/decision-support-in-medicine/pediatrics/cat-scratch-disease/

Necrotizing Ulcerative Gingivitis

“Trench Mouth”

- Bacteria: *Treponema spp.*, *Fusobacterium spp.*, *Selenomonas spp.*, and *Prevotella intermedia*
- Clinical presentation
 - Poor oral hygiene, smoking, lack of sleep, stress, malnutrition, and immunocompromised health status (HIV/AIDS)
 - Gingiva appears red, ulcerated, and necrotic
 - “Punched-out” appearance of the interdental papillae
 - Strong “fetid” odor from the oral cavity
 - Necrotizing ulcerative gingivitis  Necrotizing Ulcerative Periodontitis
 - Gingivitis changes to periodontitis when there is evidence of loss of gingival attachment and bone loss.

Necrotizing Ulcerative Gingivitis



Syphilis (Lues)

- Caused by *Treponema pallidum*
- Modes of Transmission
 - Sexual Contact
 - Mother to fetus
- Three stages
 - Primary
 - Secondary
 - Tertiary

Primary Syphilis

- Occurs at the site of inoculation
- **Chancre** appears 3-90 days after initial infection
- Lesion heals 3-8 weeks without treatment
- Oral lesions
 - Painless ulceration or vascular proliferation on the tongue, lips, palate, gingiva or tonsils

Primary Syphilis



Fig 1. Smith et al. Head Neck Pathol. 2021 Sep;15(3):787-795.



Courtesy of Oral Pathology Inc. LLC

Secondary Syphilis

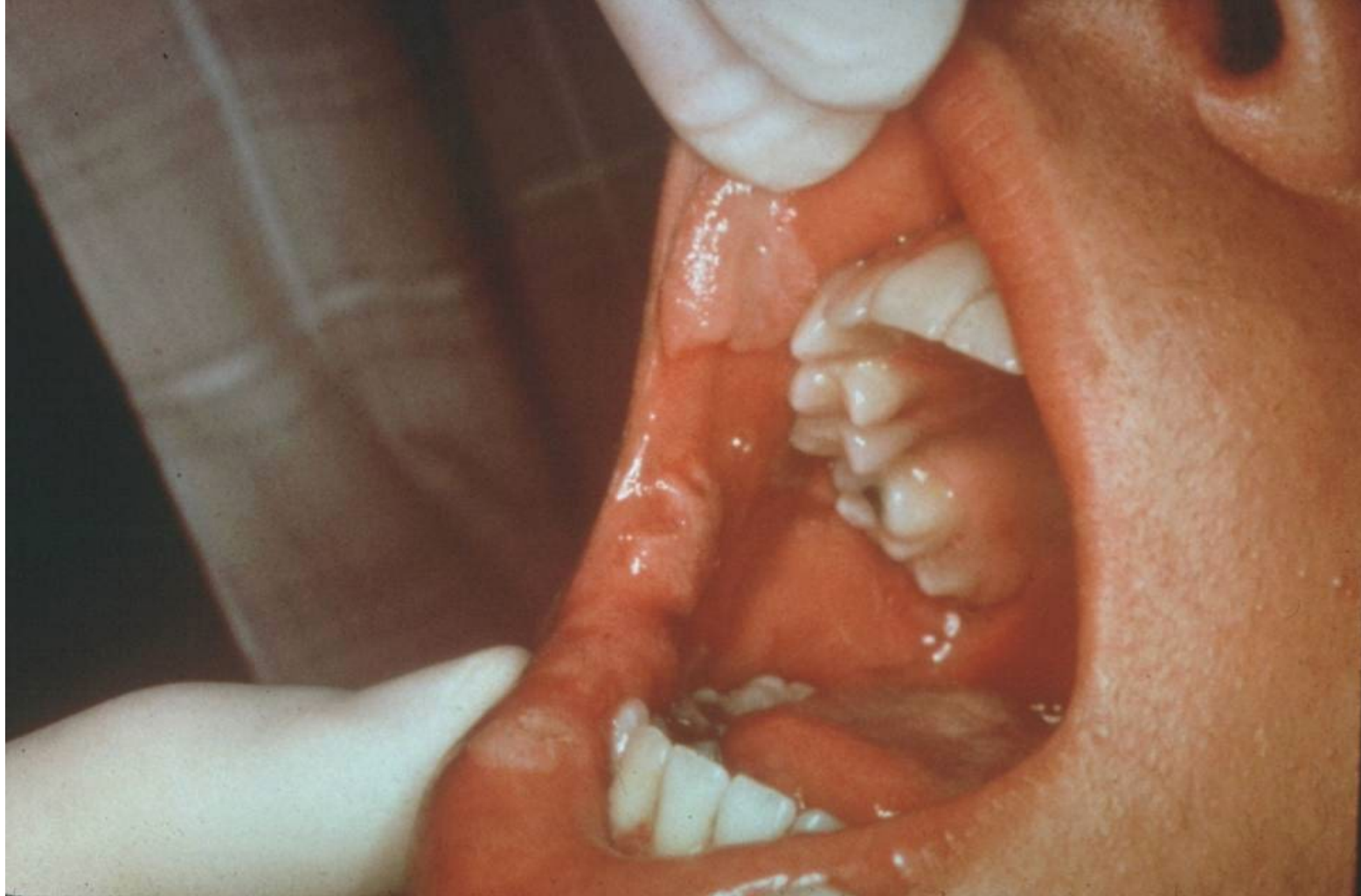
- Lesions occur 4-10 weeks after the initial infection
- Systemic symptoms
 - Painless lymphadenopathy
 - Diffuse, painless, maculopapular rash involving the palmar/plantar areas
 - Mucosal lesions:
 - Mucous patches-whitish areas of mucosa involving the lips, tongue, buccal mucosa and palate
 - Split papule- white papules with a central linear erosion
 - Condyloma lata- well demarcated hypertrophic plaques or papules

Secondary Syphilis – Maculopapular Rash



Figure 1. www.hiv.uw.edu

Secondary Syphilis – Mucous Patches



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Secondary Syphilis

Split Papule



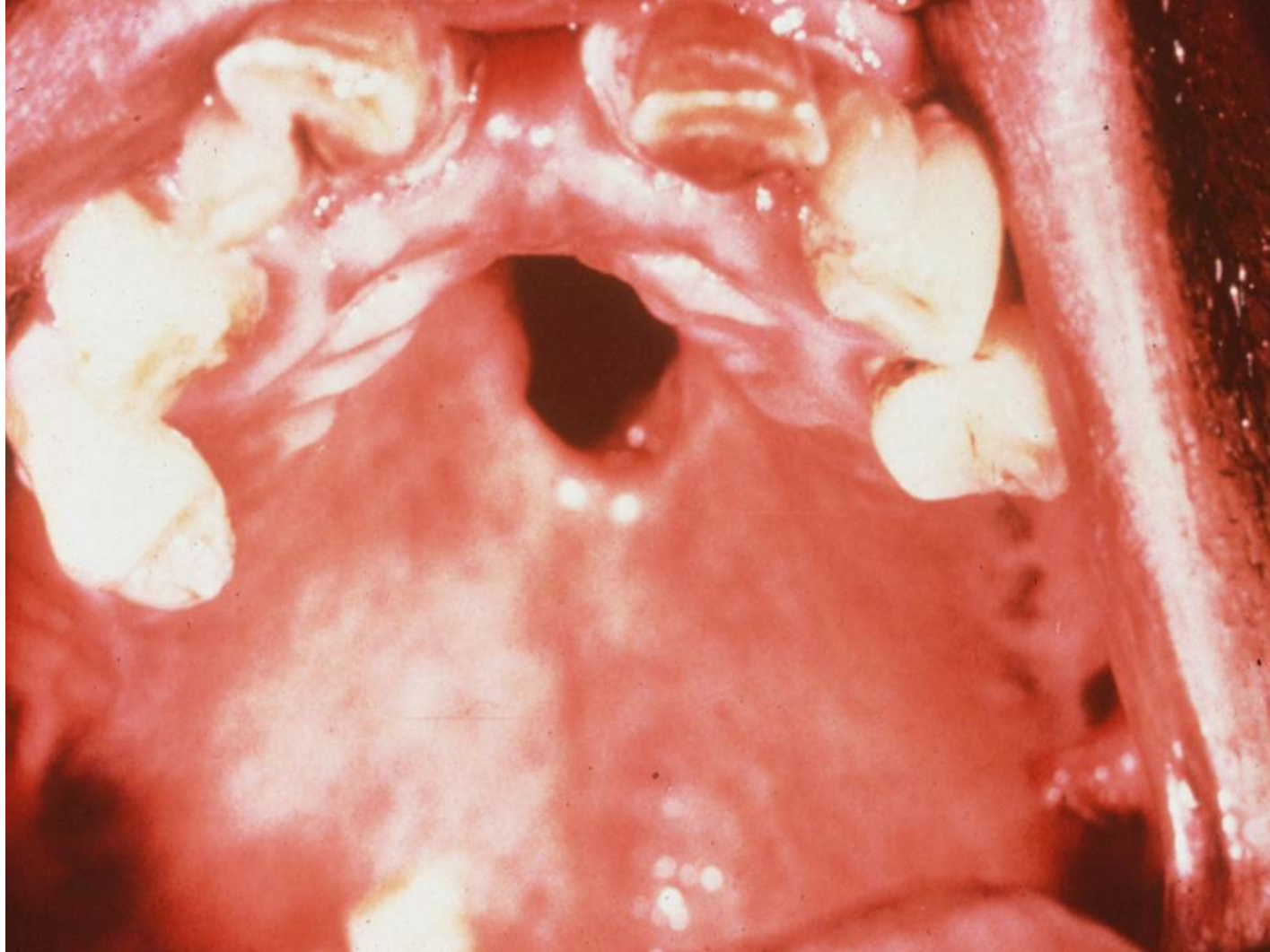
Condyloma lata



Tertiary Syphilis

- Occurs after period of latency
- Most serious complications
 - Psychosis, Dementia, Death
 - Aortic aneurysm
- Gumma
 - Indurated, nodular, ulcerated lesion resulting in tissue destruction
 - Palatal perforation
 - Granulomatous inflammation

Tertiary Syphilis - Gumma



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Tertiary Syphilis – Luetic Glossitis

Luetic Glossitis

- Loss of papilla on the dorsal surface of the tongue
- White discoloration
- Constriction of the tongue (atrophy)



Courtesy of Oral Pathology Inc. LLC

Congenital Syphilis

- Hutchinson's Triad
 - Hutchinson's Teeth
 - Incisors
 - Mulberry molars (Moon's molars)
 - Ocular interstitial keratitis
 - Eighth nerve deafness

Hutchinson's Incisors - "Screw-driver teeth"



Fig. 2 Freiman et al. J AM ACAD DERMATOL Feb. 2009

Mulberry Molars

Abnormal appearance of the cusp on the occlusal table of the molars.



Fig. 3 Freiman et al. J AM ACAD DERMATOL Feb. 2009

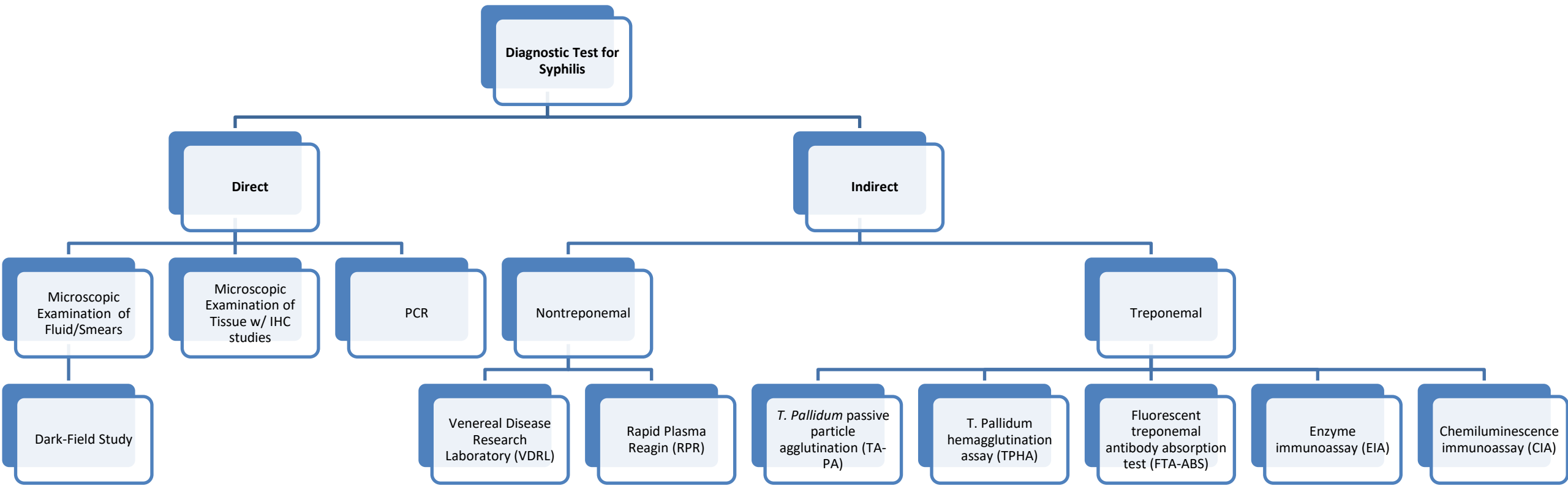
Interstitial Keratitis

Opacification of the corneal surface of the eye.

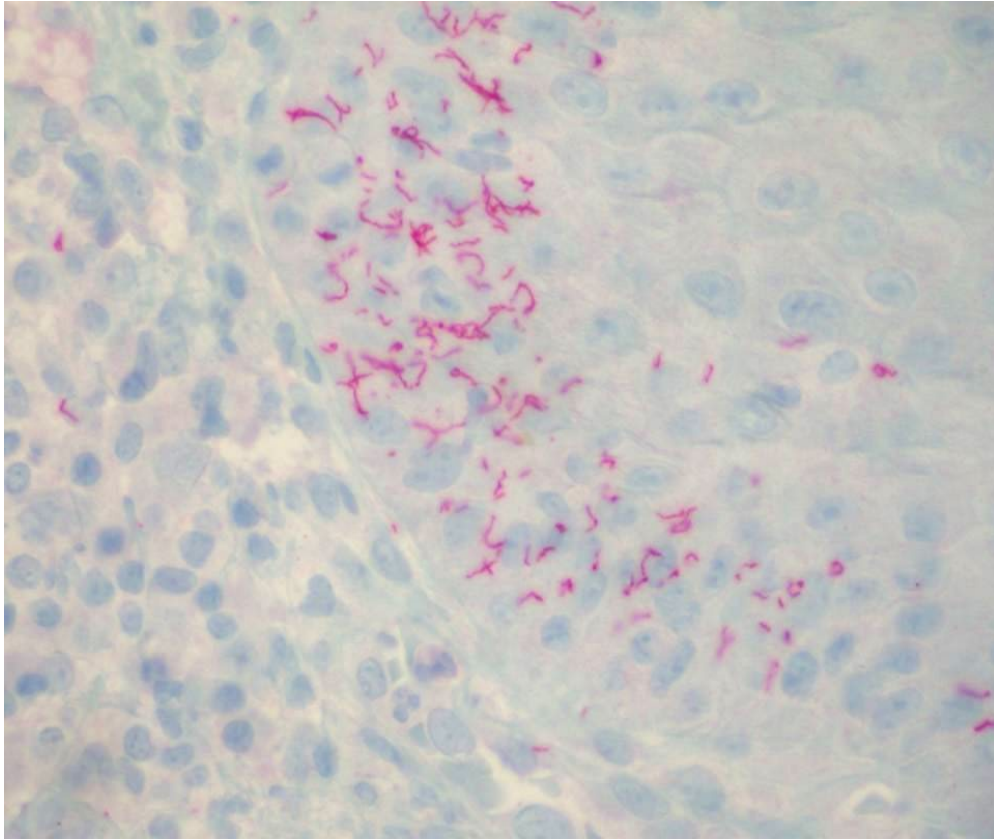
- Loss of vision in the affected eye(s)



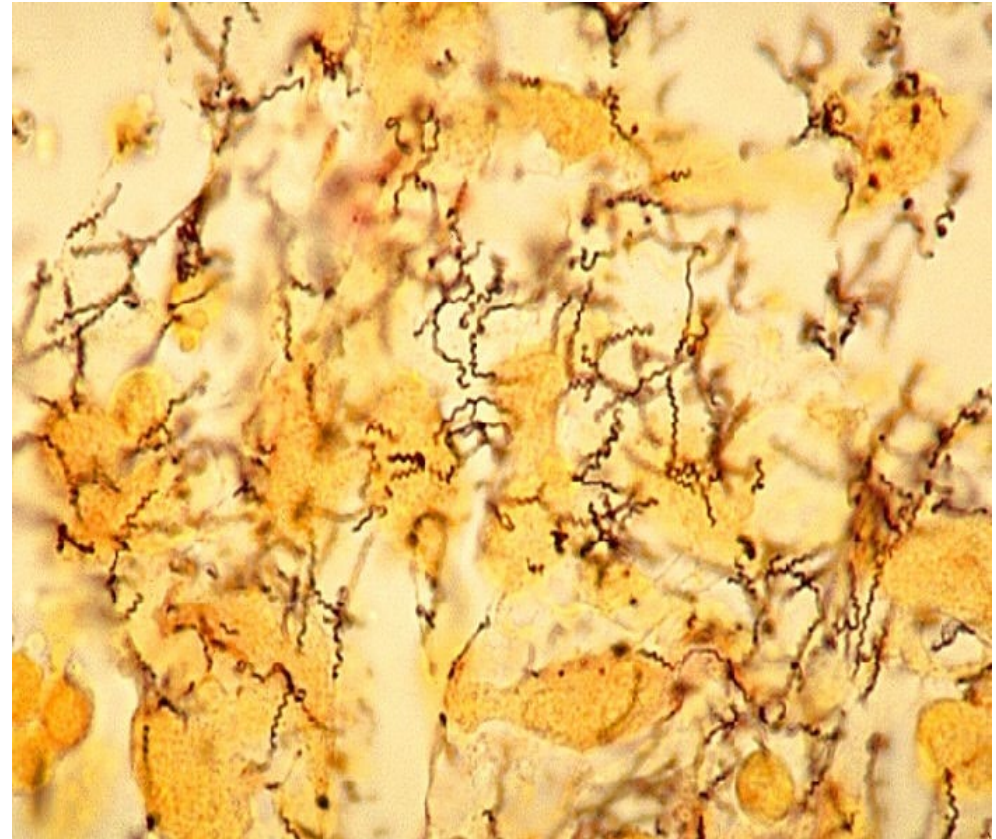
Public Health Image Library, Centers for Disease Control and Prevention. Photo by Susan Lindsley.



Direct Syphilis Diagnostic Test



Tissue with immunohistochemical staining highlighting *T. pallidum* organisms



Darkfield staining of a smear sample showing the “corkscrew” shaped organisms.

Courtesy of Oral Pathology Inc. LLC



The Tuskegee Experiment: A Failure of Science

Journal article marks 50 years since
unethical study uncovered.

AJRCCM May 2022



THE GUATEMALA INOCULATION EXPERIMENTS

 **ORIGINS**
Current Events in Historical Perspective

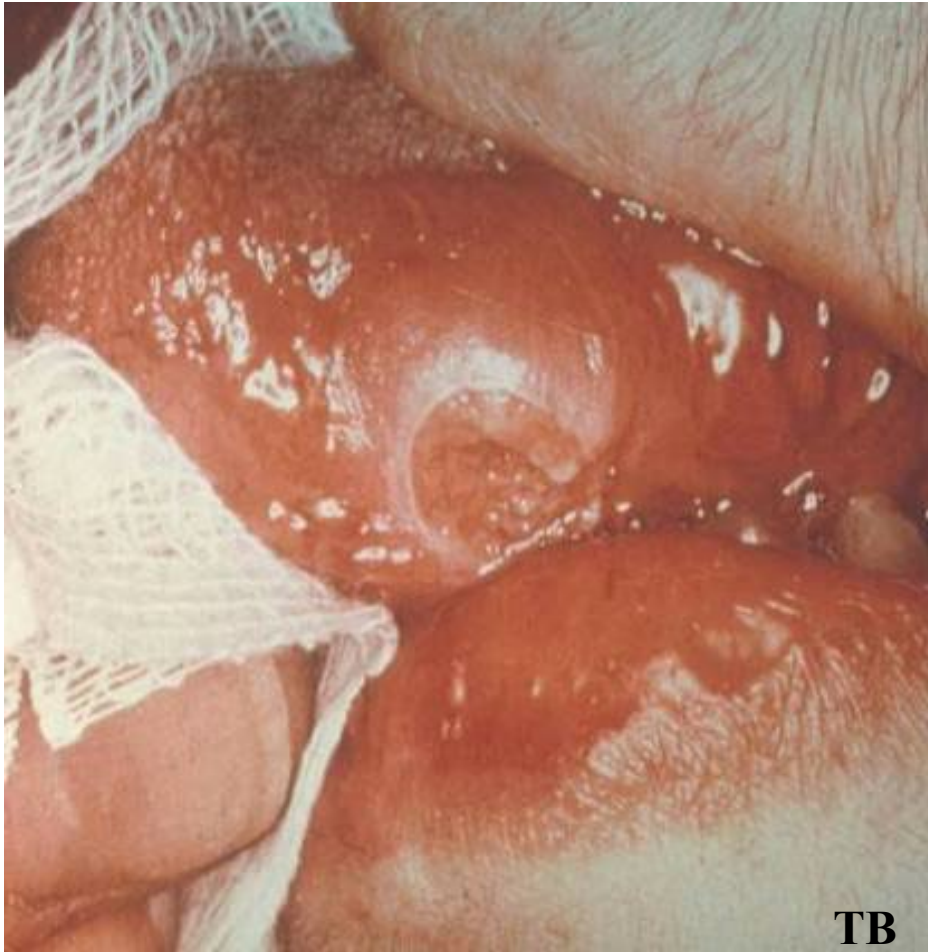
Tuberculosis

- Caused by the acid-fast bacteria *Mycobacterium tuberculosis*
- Airborne transmission via respiratory droplets
- Primary- occurs in previously unexposed people
 - Presents as a fibrocalcified nodule in the lungs
- Secondary- reactivation of the active disease

Tuberculosis

- Low grade fever, weight loss, and night sweats
- Productive cough, hemoptysis and chest pain
- Cervical lymphadenopathy
- Oral lesions
 - Chronic painless ulcer
 - Nodular and granular masses of the soft tissues

Tuberculosis



Tuberculosis- Scrofula

- AKA Cervical tuberculoid lymphadenitis
- Mycobacterial infection
- Caused by consumption of contaminated milk
- Enlargement of the oropharyngeal and cervical lymph nodes
- Sinus tracts through the overlying skin
- Pulmonary involvement is rare

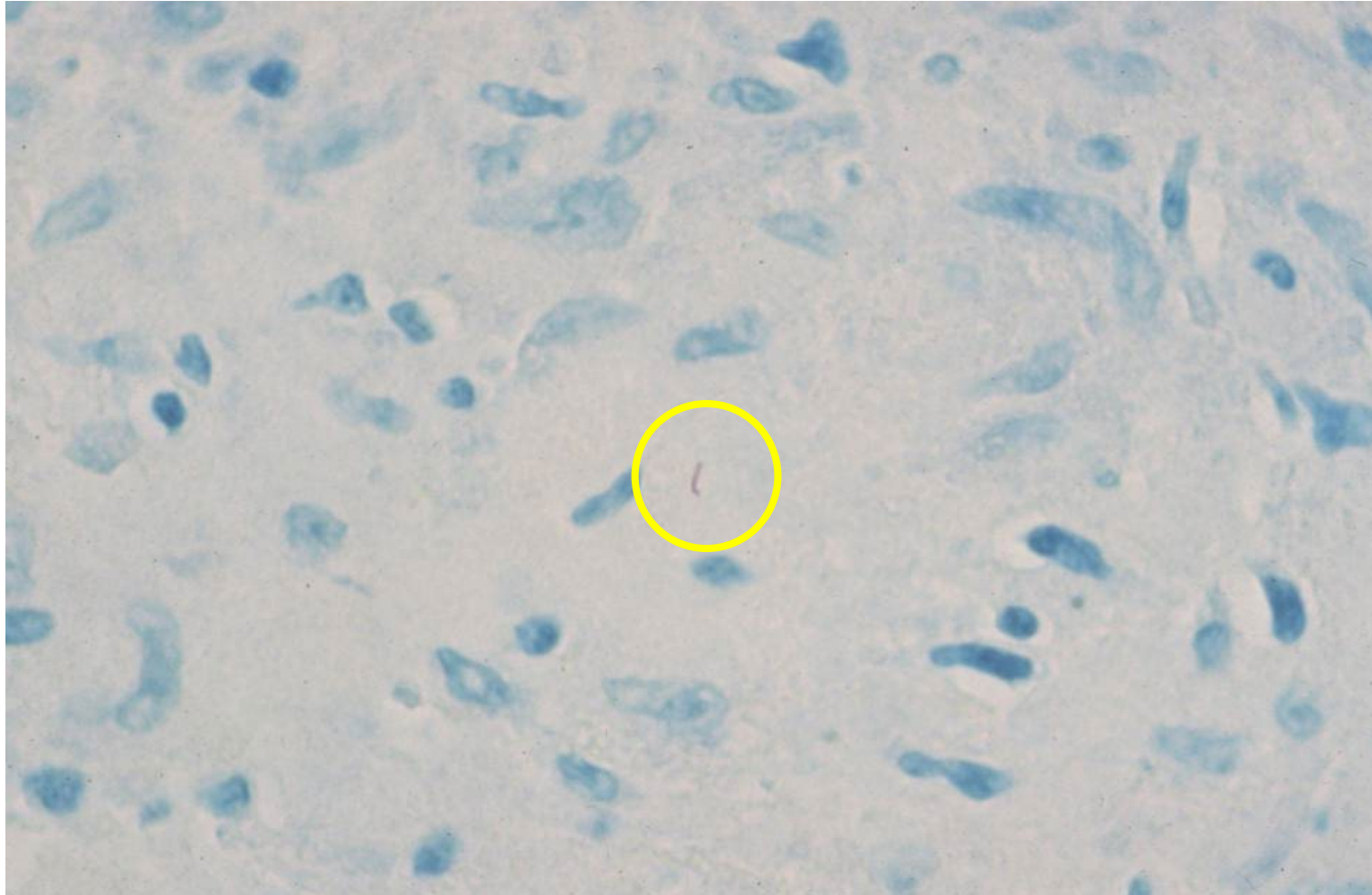
Scrofula – Lymph nodes enlargement



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Tuberculosis – Acid Fast Bacillus Stain



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Resources

